

Lumina AI Teaching System

Team Obizi

“Young Passionate IWS students”



Lumina

AI

Teach. Understand. Grow.
Your AI Teaching Companion



Personalized
Lessons



Adaptive
Teaching



Deep
Understanding



Track
Progress



AI-Powered
Learning

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Executive Summary

Lumina is a live AI teacher that transforms any learner's own material textbook chapter, lecture notes, or study guide into a structured, one-on-one lesson. Rather than waiting for students to ask questions, it teaches proactively: explaining concepts on a live whiteboard while narrating, checking understanding through in-lesson integrated practice, identifying misconceptions as they form, and adapting its instruction before allowing the learner to move on.

The breakthrough is not a better prompt or a larger language model. It is creating a new category of educational software AI that doesn't simply answer questions or deliver content, but continuously teaches, verifies understanding, and guides every learner through lesson as a personal tutor would. Lumina embeds pedagogy directly into software. This architecture makes teaching behaviour consistent, explainable, and independent of the specific language model answering each request.

That design also makes Lumina economically scalable. High-frequency learner interactions are served by a low-latency model running on AMD Instinct hardware, while computationally intensive lesson planning is routed to a higher-capability reasoning model on AMD-powered infrastructure through a unified gateway. By separating pedagogical logic from model execution, Lumina delivers responsive, personalized instruction while keeping the latency and operating costs required for a subscription business.

PROBLEM STATEMENT

Students today are drowning in study material and while starving for understanding. Access to content has never been greater than endless videos, courses, PDFs, AI chatbots, homework solvers and yet millions of learners still fail to master what they're taught. The bottleneck was never availability. It's responsiveness.

In 1984, Benjamin Bloom found that students who received one-on-one mastery tutoring outperformed 98% of students in conventional classrooms a gap so large it became known as the "2 Sigma Problem." His conclusion was the most effective teaching method ever measured, and the least scalable. No economy can staff a personal tutor for every student.

Forty years later, that constraint hasn't been solved and it's been reproduced at scale in three different forms. Classrooms force one teacher to guide many students without checking each one's understanding before moving on. Prerecorded online courses deliver one fixed sequence with no way to know if it landed. General AI chatbots and homework solvers answer whatever's asked but hand the entire burden of directing learning back to the student, who must know what to ask, when, and whether they understand the answer.

The failure mode is identical across all three, content built for anyone at any level, where gaps compound silently. A student who never understood fractions struggles with algebra. Receiving an answer gets mistaken for understanding it.

Why Now?

Bloom published this finding in 1984. The reason no one built this in 1995, or 2015, isn't that the insight was missing, it's that the technology to act on it didn't exist. Continuous one-on-one instruction at scale requires a system that can reason about a specific learner's specific confusion in real time, hold state across an interruption without losing the thread, and do it cheaply enough to run per-turn, for every learner, indefinitely. That combination of affordable, low-latency reasoning inference plus stateful, structured conversation has only become viable in the last few years. Lumina exists now because the constraint just moved from "impossible" to "an infrastructure problem," and infrastructure problems can be engineered around.

The Lumina Teaching System

Lumina is an AI teaching system designed to deliver personalized, interactive, and responsive real time learning. Instead of acting as a chatbot that simply answers questions, Lumina behaves like a real teacher. It plans lessons, explains concepts, checks understanding throughout the lesson, adapts when a student is struggling, and only moves forward once the student has demonstrated understanding. By combining artificial intelligence with structured teaching methods, Lumina transforms existing educational content into a personalized learning experience that helps students achieve genuine understanding.

The proposed solution doesn't add more educational content to a world already full of it, it turns content that already exists (a teacher's curriculum, a textbook, a set of lecture notes) into a live, responsive teaching process.

What Lumina Actually Does

A learner uploads a chapter of their calculus textbook. Lumina reads it, breaks it into a sequence of lessons, and before ever speaking a word to the student, it creates a lesson plan of what it is going to teach, in what order, and where students in this specific topic typically get stuck so it anticipate questions that might arise.

The lesson begins. Lumina explains limits, writing on a live board while narrating, the way a teacher writes while still talking. It doesn't wait for the student to ask for a check-in, it assesses understanding through practice problems built into the lesson itself. The student answers one and gets it subtly wrong in a way that suggests they've confused a limit with a value the function never actually reaches.

Lumina doesn't just correct the answer. It identifies that the error traces back to a misunderstanding of continuity a concept from two steps earlier pauses the current explanation, re-teaches specifically that point using a different approach, confirms the student now gets it, and then resumes the original lesson exactly where it left off. No restart, no repeated content, no separate "help" mode the student must switch into.

The student never leaves the lesson. They never open a tab to search for a YouTube explainer. And the gap that would have silently compounded into a harder misunderstanding next month gets caught and closed the moment it forms.

Technical Architecture

Every one of the behaviours above is easy to describe. Making them reliable is a different problem, and it's the one Lumina's architecture solves. Four questions a technical reviewer should ask answered plainly:

- **How is a lesson generated?**
Not as a single "write me a lesson" prompt. Lumina's lesson engine enforces a strict four-level hierarchy, Lesson → Sections → Steps → Presentation Events with pedagogical rules built into the structure itself, not left to a model's discretion. A concept can't be introduced without a worked example and a practice section attached; that rule is enforced in code, so lesson quality doesn't degrade as usage scales the way ad-hoc prompting does.
- **How does it resume mid-lesson without losing the thread?**
Because a "step", the smallest unit in that hierarchy is deliberately atomic, one teaching action, small enough that an interruption can resume from that exact point. Lumina isn't stuffing a growing transcript into context and hoping the model infers where it left off. It's tracking an explicit state object, which step is active, which teaching approaches have already been tried, whether understanding was confirmed, so resume is exact, not approximate, no matter how long the lesson has run.
- **How does it know where a misconception is?**
A single LLM call guessing from a wrong answer is unreliable and gives you no way to guarantee it doesn't just repeat the same explanation in different words. Lumina's answer engine instead runs a fixed escalation ladder of teaching approaches, grades the student's response to its own probes at each step, and if understanding still isn't confirmed, probes backward specifically to locate where confusion began, rather than escalating blindly.
- **Why isn't this just prompting GPT?**
Because none of the above are prompting decisions, they're structural ones. The lesson hierarchy, the atomic step sizing, the escalation ladder, and the state tracking all exist in code, independent of which model answers any individual call. That's also why Lumina routes different calls to different model tiers through a single LLM gateway, large, once-per-lesson planning calls that can afford to think longer go to a reasoning tier model; the high-frequency, per-turn calls that fire on every learner interaction and directly determine both responsiveness and cost, go to a faster, cheaper tier. A prompt wrapper product has no such distinction, because it has no structure to route around.

AMD Infrastructure

The per-turn tier is the part of the system that decides whether a subscription business survives contact with real usage, every understanding check, every acknowledgment, every graded response fires on every single learner turn, for every learner, continuously. Lesson planning happens once per lesson and can afford to think. Per-turn checking happens constantly and can't so it's the per-turn tier's cost and latency, not planning quality, that determines whether the unit economics of a subscription business hold up on a scale.

Lumina's LLM gateway is the only place in the system that knows which hardware answers a given call. The lesson engine, streaming engine, and answer engine never know or care they request a logical tier ("reasoning" or "fast") and the gateway resolves it. That's a deliberate architectural choice, if we needed to move off AMD Hardware tomorrow, the change is one routing configuration inside the gateway, not a rewrite of the application.

Two tiers, both running on AMD Instinct hardware:

Fast tier; Qwen3-14B, self-hosted on our own AMD Instinct GPU instance via vLLM. This is the tier answer engine calls on every single learner turn checking an answer, grading a response to a probe, deciding whether to escalate to a different teaching approach. It must return in a few hundred milliseconds or the "live teacher" illusion breaks, and it must be cheap enough to run per-turn, for every concurrent learner, without the subscription model collapsing.

Reasoning tier; Qwen 3.7 Plus, hosted on Fireworks AI's AMD Instinct MI300X infrastructure. This handles lesson planning and the harder in-lesson explanations, where we deliberately let the model think longer using Fireworks' reasoning effort parameter, a tuneable control exposed by Qwen 3.7 Plus's hybrid-reasoning mode. Planning happens once per lesson, so the extra latency is worth spending on quality output.

Both tiers deliberately run the same model family, Qwen's hybrid-reasoning checkpoints rather than mixing model families across tiers. That was an intentional design decision, not a coincidence a lesson planned by one model family and taught turn-by-turn by a different one risks a mismatch in how the two explain and phrase things, which a learner would feel as inconsistent teaching even when models are individually strong. Keeping one model family across both tiers, differing only in size and in where each instance runs, is what keeps that inconsistency from ever surfacing in the actual lesson.

This wasn't the starting design. Lumina originally routed everything through OpenAI's gpt-4omini. It worked, but the fast tier's per-turn latency was the single biggest constraint on how responsive the "live teacher" experience could feel, every answer check added a delay a real teacher would never add. Migrating the fast tier to a self-hosted Qwen3-14B instance on AMD Instinct hardware is what took per-turn response time from a noticeable pause to something close to real-time. That's why the fast tier, the one that fires on every learner interaction, for every learner, continuously, and therefore dominates both latency and operating cost, is the one we chose to fully own on AMD infrastructure rather than lease from a general-purpose hosted API.

Design Principles

- Teaching is stateful
- Pedagogy lives in software, not prompts.
- Lessons are hierarchical
- Every step is resumable
- Understanding is verified before progression
- Infrastructure is abstracted from teaching

Market Opportunity

The global AI-in-education market is estimated at roughly \$10 billion in 2026, growing 25–40%+ annually, with projections reaching \$30–70 billion by the early 2030s. That growth is driven by two converging trends, rising demand for personalized learning, and AI finally becoming capable of real-time reasoning and natural interaction.

Most existing investment has targeted access, more content, more courses, more chatbots, not delivery quality. That leaves an open category, AI teaching systems that actively guide learners, continuously verify understanding, and adapt instruction as it happens. Lumina is built to be the first system in that category to scale personalized instruction to millions of learners rather than one classroom or one chat session at a time.

Go-to-Market Strategy

Lumina is launching direct-to-learner into high school and university STEM mathematics, physics and engineering. These subjects are progressively structured, every concept depends on mastery of the last, so an undetected gap doesn't just persist, it actively blocks the next unit. That makes STEM the clearest proof ground for the core claim: continuously verified understanding, not just delivered content, is what determines whether learning sticks.

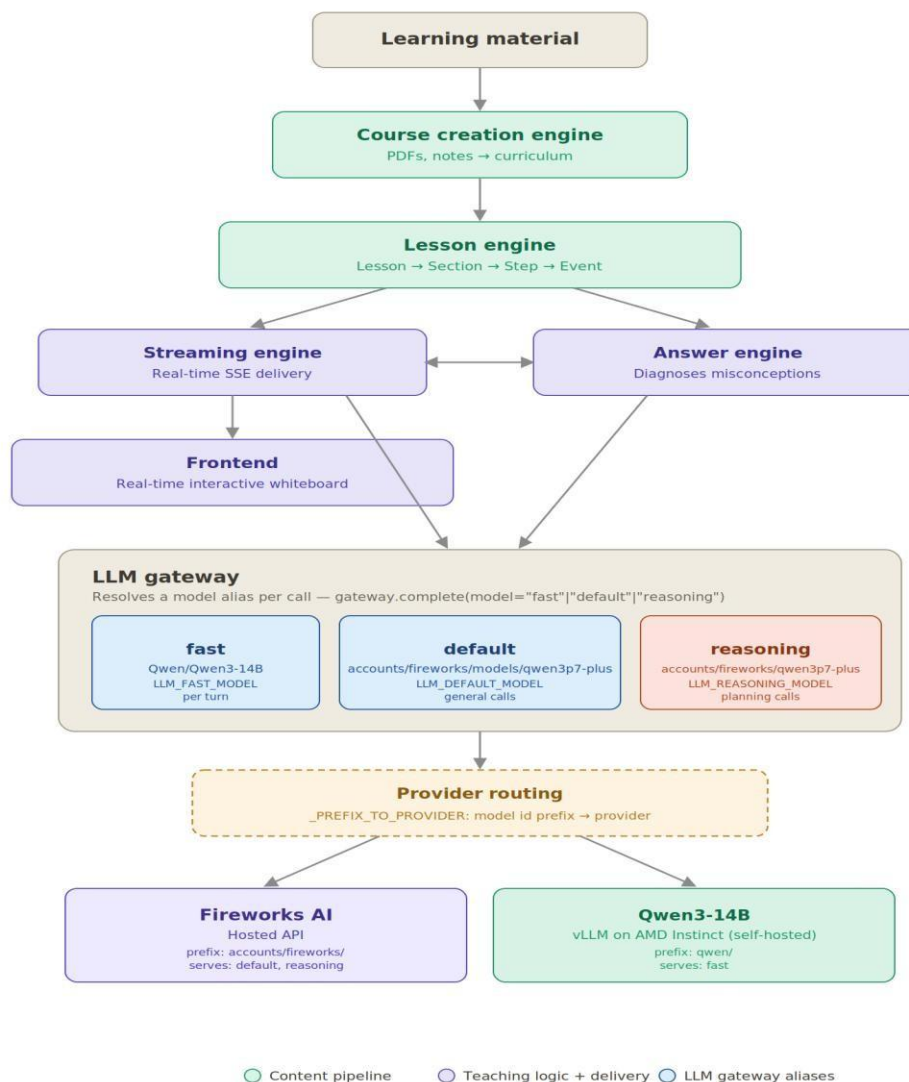
The initial product replaces static, one-size-fits-all textbooks and prerecorded courses with a subscription-based teaching experience calibrated to the individual learner the same curriculum, taught adaptively to the person using it. Going direct-to-learner first allows fast iteration, direct feedback, and measurable outcomes before Lumina expands into institutional partnerships with schools and universities.

Students already spend billions annually on textbooks, tutoring, online courses, and exam preparation. Lumina consolidates those fragmented expenses into a single adaptive teaching subscription.

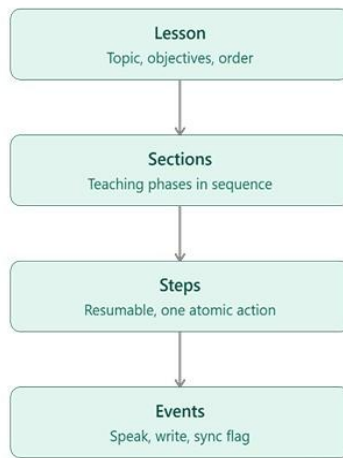
SYSTEM ARCHITECTURE

Lumina is a live AI teacher, not another AI teaching platform in the crowded sense of that word the distinction matters because everything above describes the behaviour of a teacher rather than a content platform, it prepares before a lesson runs, notices confusion as it happens, teaches to the specific gap rather than repeating itself, allows mid-lesson questions, and never lets a student move forward without actually understanding what came before. Today Lumina already demonstrates automated curriculum generation, structured lesson planning, live teaching, misconception diagnosis, adaptive reteaching, interruption recovery, and AMD powered multi-tier inference. Our next milestone is validating learning outcomes with real students before expanding into institutional deployments.

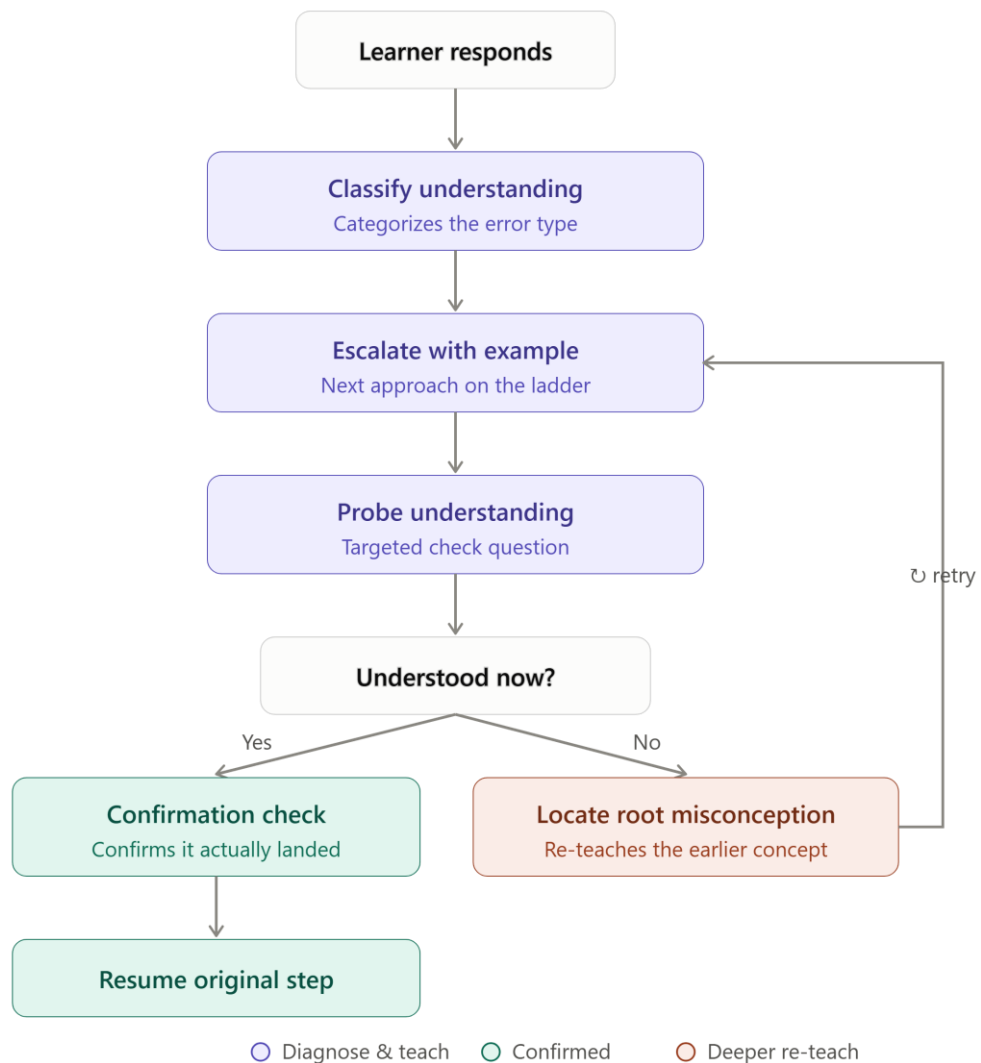
1. Overall architecture



2. Lesson hierarchy



3. Misconception loop



4. LLM gateway

